

Representation-Steered Incremental Adapter-Tuning for Class-Incremental Learning with Pre-Trained Models

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Abstract

Class-Incremental Learning (CIL) aims to develop models to continuously learn new classes without forgetting learned old ones. Recent advances combine pre-trained models with parameter-efficient fine-tuning, achieving promising results. However, these approaches typically allocate new trainable parameters for each task, causing the model size to grow linearly with task number. Moreover, they lack explicit mechanisms to structure a coherent and discriminative representation space across tasks. To address these limitations, we propose Representation-Steered Incremental Adapter Tuning (RSIAT). RSIAT maintains a single shared adapter for all tasks, eliminating parameter growth during incremental learning. In the base task, we introduce a representation-steering loss that enhances discriminative feature learning while facilitating future task adaptation. During incremental tasks, a residual autoencoder-based projector aligns feature distributions between old and new tasks, preserving representation consistency without over-constraining the shared adapter. Extensive experiments on six CIL benchmarks demonstrate that RSIAT significantly outperforms state-of-the-art methods in both performance and parameter efficiency, achieving superior stability-plasticity trade-offs with minimal trainable parameters. Code is available at <https://github.com/zjrjrz/RSIAT>.

1. Introduction

In traditional deep learning, models can access all data for unified training. However, real-world data often arrives in streaming formats. Therefore, real-world vision systems are required to continuously learn new classes without for-

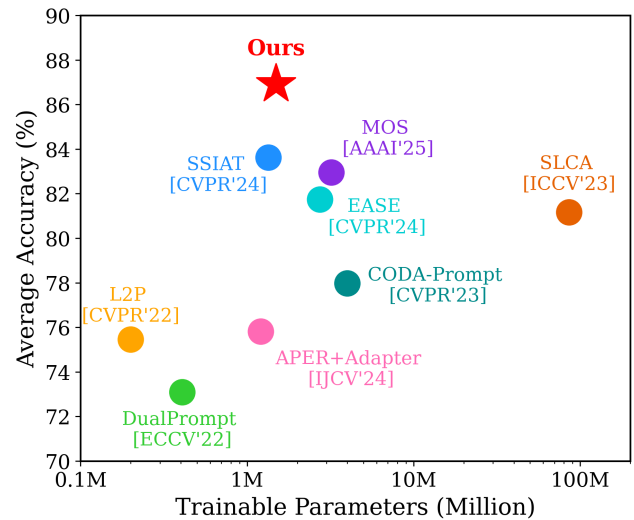


Figure 1. Parameter-Accuracy comparison on ImageNet-R (B0I20). Each point shows the average accuracy (%) versus the number of total trainable parameters (Million). All methods are based on the same pre-trained model (ViT-B/16-IN21K). Our method (RSIAT, red star) uses a constant-size adapter, avoiding task-specific module growth while further improving accuracy.

getting old ones [2], known as Class-Incremental Learning (CIL). In CIL, a major challenge is addressing catastrophic forgetting [8], and various methods [18, 23, 24] have been developed to alleviate this issue.

Traditional CIL methods can be broadly categorized into replay-based [3, 4, 14, 32, 64], regularization-based [1, 15, 34, 58], and isolation-based methods [26, 27, 33]. However, with the widespread adoption of large Pre-Trained Models (PTMs) [11], research has gradually shifted from “training from scratch” to Parameter-Efficient Fine-Tuning (PEFT) [16, 21, 22, 42–44]. Unlike updating the entire backbone network, PEFT adapts small auxiliary training modules (e.g., prompts [17], adapters [5]) to the frozen

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backbone. PTM-based CIL methods have greatly advanced performance but also exposed two fundamental issues. First, most approaches incrementally accumulate task-specific modules as new tasks arrive. When a new task is introduced, the model typically appends additional prompts or adapters that must be retrieved during inference. This design causes a linear increase in trainable parameters, and mis-selection or interference among modules can significantly degrade performance [10]. Second, despite the strong generalization capability of PTMs, current methods offer limited guidance for constructing a well-structured and task-consistent representation space. Without explicit regularization [49, 50], newly learned tasks induce feature drift in previously learned categories, manifesting as prototype shifts [7, 45], blurred decision boundaries [56], and reduced representation stability [62].

Motivated by these issues, we posit that an ideal CIL system should satisfy two key requirements: (i) parameter growth should be limited or ideally constant throughout training; and (ii) the feature space should remain compact within classes, sufficiently separable across classes, and aligned between historical and new tasks. To this end, we propose Representation-Steered Incremental Adapter Tuning (RSIAT), a PTM-based CIL framework designed from a representation-first perspective. RSIAT shares a single adapter across all tasks, preventing parameter bloat and eliminating the need for instance-level module retrieval. In the base task, we introduce a representation-steering loss that encourages intra-class compactness and inter-class separability while preserving PTM transferability. This loss promotes discriminative structures, such as well-defined decision boundaries and prototype compactness, enabling subsequent tasks to build on a robust and adaptive feature space. In incremental tasks, a lightweight residual autoencoder-based projection maps new task features to the representation space established by previous tasks. This residual alignment mechanism suppresses prototype drift, maintains clear decision boundaries, and improves representation stability without over-regularizing the shared adapter. As illustrated in Fig. 1, RSIAT achieves higher average accuracy with fewer trainable parameters than existing PTM-based CIL methods, striking a favorable balance between accuracy and parameter efficiency.

In summary, our main contributions are:

- We propose RSIAT, a PTM-based CIL framework that shares a single adapter across all tasks, preventing parameter growth while enabling efficient incremental learning.
- We introduce a representation-steering loss in the base task and a residual autoencoder-based projection in incremental tasks to enforce compact, discriminative, and aligned feature spaces, mitigating prototype drift and preserving decision boundaries.
- We demonstrate that RSIAT achieves state-of-the-art

accuracy-efficiency trade-offs on six CIL benchmarks, outperforming existing methods while keeping trainable parameters constant.

2. Related Work

2.1. Class-Incremental Learning (CIL)

Class-Incremental Learning (CIL) extends continual learning by requiring models to continuously learn new classes while maintaining performance across all previously encountered classes under a unified classifier [61]. Existing CIL approaches can be broadly categorized into three paradigms. Replay-based methods [3, 4, 32] mitigate forgetting by retaining exemplars or synthetic samples from past tasks and interleaving them with new data during training. These methods often design sophisticated sample selection or memory management strategies to balance knowledge retention and efficiency [54, 55]. Regularization-based methods [1, 15, 58] constrain updates to parameters deemed important for previously learned classes, thereby preserving historical knowledge without explicit data replay [52, 53, 60]. Isolation-based methods [26, 27, 33] allocate distinct model capacities for different tasks through pruning, masking, or gating, effectively mitigating interference among tasks. To further enhance representational flexibility, network expansion methods [40, 47, 65] dynamically increase the model’s capacity, expanding the feature extractor as new tasks arrive to accommodate emerging knowledge.

Despite their effectiveness, most of these approaches rely on extensive retraining or growing architectures, motivating the shift toward parameter-efficient adaptation that leverages frozen pre-trained models as stable knowledge bases for scalable CIL.

2.2. Pre-Trained Model-Based CIL

Leveraging the strong generalization ability of Pre-Trained Models (PTMs) [51], PTM-based Class-Incremental Learning (CIL) has recently emerged as a dominant paradigm [41, 46]. These methods shift the focus from training networks from scratch to adapting powerful PTMs through parameter-efficient fine-tuning (PEFT), aiming to retain generalization while continually learning new classes.

Early explorations adopted fine-tuning and projection-based strategies. SLCA [62] stabilizes representations by coupling a slowly updated backbone with a fast-learning classifier refined via feature generation. RanPAC [28] offers a training-free alternative by freezing the PTM and accumulating class prototypes through a random-projection layer, providing a strong rehearsal-free baseline. Prompt-based tuning methods then advanced PTM utilization by introducing learnable, task-adaptive prompts. L2P [44] employs an instance-conditioned prompt pool for dynamic retrieval,

while DualPrompt [43] separates prompts into general and expert components to capture both task-invariant and task-specific knowledge. CODA-Prompt [35] further enhances flexibility through modular prompt composition guided by attention mechanisms. Recent developments increasingly adopt adapter-based tuning to better balance adaptability and stability. APER+Adapter [66] combines frozen PTMs with prototype-based classification and branch merging to enhance generalization. LAE [9] introduces EMA-based calibration and integrates PEFT for adaptive prediction. EASE [67] learns lightweight adapters to span a scalable subspace and employs semantic prototype completion to mitigate cross-task interference. MOS [37] alleviates both parameter- and retrieval-level forgetting via adapter merging and self-refined retrieval, while SSIAT [38] maintains a shared adapter updated with semantic drift awareness and retrain the classifier through prototype sampling to reinforce stability-plasticity trade-offs.

Despite these advances, most PTM-based CIL methods focus on task-specific prompts or adapters and lack explicit mechanisms to guide representation structuring across tasks. In contrast, RSIAT bridges this gap by maintaining a single shared adapter and explicitly steering representations throughout incremental learning, achieving both compactness and cross-task coherence.

3. Preliminaries

3.1. Class-Incremental Learning

We introduce the definition of CIL. Data arrive in tasks $\{\mathcal{D}_t\}_{t=1}^T$, where $\mathcal{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^{n_t}$ and $y_i^t \in \mathcal{Y}_t$. Label spaces are disjoint across tasks, $\mathcal{Y}_t \cap \mathcal{Y}_{t'} = \emptyset$ for $t \neq t'$. At task t , the model can only access $\mathcal{D}_t$ during training, and is evaluated on the cumulative label space $\mathcal{Y}_{1:t} = \mathcal{Y}_1 \cup \mathcal{Y}_2 \cup \dots \cup \mathcal{Y}_t$. We hope to find a model $\mathbf{f}(\mathbf{x}) : X \rightarrow \mathcal{Y}_b$ that minimizes the expected risk:

$$\mathbf{f}^* = \arg \min_{\mathbf{f} \in \mathcal{H}} \mathbb{E}_{(x,y) \sim \mathcal{D}_1 \cup \dots \cup \mathcal{D}_t} \mathbb{I}(y \neq \mathbf{f}(x)), \quad (1)$$

where $\mathcal{H}$ is the hypothesis space and $\mathbb{I}(\cdot)$ denotes the indicator function. Following typical PTM-based CIL works [67], we decouple a pre-trained backbone into a frozen feature extractor $\phi : \mathbb{R}^D \rightarrow \mathbb{R}^d$ and a classifier $W \in \mathbb{R}^{d \times |\mathcal{Y}_{1:t}|}$:

$$\mathbf{f}(x) = W^\top \phi(x), \quad W = [w_1, \dots, w_{|\mathcal{Y}_{1:t}|}]. \quad (2)$$

In PTM-based CIL, PEFT adapts a limited number of trainable modules (e.g., adapters, prompts) while keeping the pre-trained model frozen, which preserves prior knowledge and ensures parameter-efficient learning.

3.2. Baseline

Incremental adapter tuning. Our baseline builds directly upon SSIAT [38], which performs incremental adapter tun-

ing with a shared adapter A , while keeping the PTM backbone frozen. Given a batch $\mathcal{B}_t = \{(x_i, y_i)\}_t$ of samples from the new class set $\mathcal{Y}_t$, they extract embeddings $f_i^t = \phi(x_i; A^t) \in \mathbb{R}^d$ and the ℓ_2 -normalized features $\hat{z}_i^t = f_i^t / \|f_i^t\|$. Note that, during task t , only the adapter A^t and the classifier are trainable. Let the classifier weights be $W^t = [\mathbf{w}_c]_{c \in \mathcal{Y}_t}$, $\cos \theta_i^c = \hat{\mathbf{w}}_c^\top \hat{z}_i^t$ with $\hat{\mathbf{w}}_c = \mathbf{w}_c / \|\mathbf{w}_c\|$, a cosine loss is engaged [30],

$$\mathcal{L}_{\cos}^{(t)} = -\frac{1}{|\mathcal{B}_t|} \sum_{(x_i, y_i) \in \mathcal{B}_t} \log \frac{\exp(s(\cos \theta_i^{y_i} - m))}{\exp(s(\cos \theta_i^{y_i} - m)) + \sum_{c \in \mathcal{Y}_t \setminus \{y_i\}} \exp(s \cos \theta_i^c)}, \quad (3)$$

where s and m are the fixed scale and margin hyperparameters, respectively, enforcing tighter decision boundaries.

Semantic shift. After each task, we store the prototype and covariance of every new class as frozen statistics. This adds no trainable parameters and keeps the inference-time model size unchanged. However, as the shared adapter continues to be tuned on later tasks, features of earlier classes inevitably drift. Following prior work [38, 57], we estimate this drift by comparing embeddings of the current task's data before and after adaptation, treating the shift as local variation to update older class prototypes. With updated prototypes and covariances, we model each class as a Gaussian-like distribution to synthesize features, which are then used to retrain the unified classification head. However, using only Eq. 3 during adapter tuning imposes weak geometric constraints, resulting in overlapping clusters, inspiring our representation steering.

4. Methodology

We propose a Representation-Steered Incremental Adapter Tuning (RSIAT) method, which is built upon the baseline. The framework of RSIAT is shown in Fig. 2. Its central idea is to balance stability and plasticity through a two-stage, continuous mechanism: (i) in the base task, we shape the representations to be compact within classes and well-separated between classes, allowing subsequent tasks to begin training from a more easily maintained representation structure; (ii) in the incremental task, we use a residual autoencoder (RAE) to align the evolution space, thus avoiding optimization resistance from strong regularization, while imposing moderate orthogonality constraints to prevent new features from overlapping with old prototypes. We consistently use a shared adapter to avoid parameter growth and module retrieval during inference.

4.1. Base Representation Steering

Representation steering. In the base task, the frozen PTM backbone network with shared adapter needs to be adapted

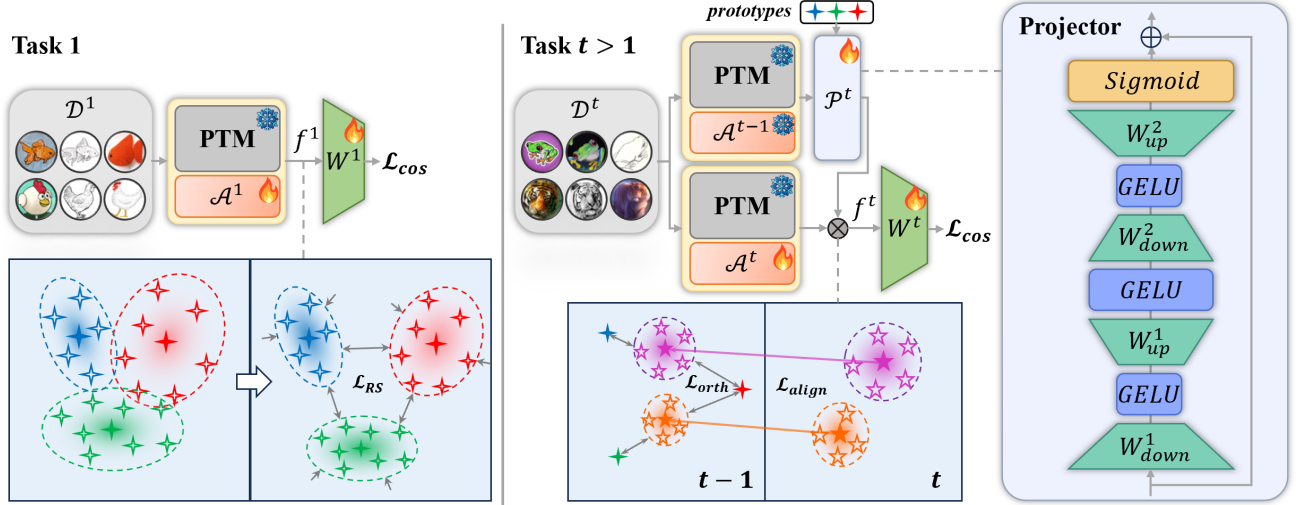


Figure 2. Overview of RSIAT. **Left:** Task 1 trains a shared adapter with a cosine classifier and a warm-up representation-steering loss $\mathcal{L}_{RS}$, which compacts within-class features and enlarges inter-class margins. **Right:** Task $t > 1$ updates the same adapter while a residual autoencoder projector P^t projects features from the previous model to the current space. The alignment loss $\mathcal{L}_{align}$ and the orthogonal loss $\mathcal{L}_{orth}$ align the feature space across tasks and steer the learning of new classes, while preserving the representations of old classes. The Projector is a lightweight autoencoder with an identity skip (residual).

to the dataset. We use the loss in Eq. 3 to train the adapter A^1 and classifier W^1 . As shown in Fig. 2, although this loss can help the model learn new classes, it cannot guide the model to construct a well-structured representation space. To actively construct a good feature space, we introduce a pairwise representation-steering (RS) loss. This loss maintains the transferability of PTM while promoting intra-class compactness and inter-class separability, enabling subsequent tasks to be trained on a feature space that is both robust and adaptive. RS loss can be decomposed into a positive loss ($\mathcal{L}_{pos}$) and a negative loss ($\mathcal{L}_{neg}$). The positive loss encourages same-class features to cluster, while the negative loss encourages different-class features to separate. Let $\hat{\mathbf{z}}_i^1$ denote the i -th normalized feature in the batch, $S_{ij} = \langle \hat{\mathbf{z}}_i^1, \hat{\mathbf{z}}_j^1 \rangle$ be the cosine similarity between samples i and j in batch $\mathcal{B}_1$, and $M_{ij}^+ = \mathbb{I}(y_i = y_j, i \neq j)$ and $M_{ij}^- = \mathbb{I}(y_i \neq y_j)$ be the positive and negative pair masks, respectively. The RS loss can be written as:

$$\begin{aligned}\mathcal{L}_{pos} &= \frac{1}{|P|} \sum_{i,j} (1 - S_{ij}) M_{ij}^+, \\ \mathcal{L}_{neg} &= \frac{1}{|N|} \sum_{i,j} (1 + S_{ij}) M_{ij}^-, \\ \mathcal{L}_{RS} &= \mathcal{L}_{pos} + \alpha \mathcal{L}_{neg},\end{aligned}\quad (4)$$

where $P = \{(i, j) \mid M_{ij}^+ = 1\}$ and $N = \{(i, j) \mid M_{ij}^- = 1\}$ collect the indices of positive and negative pairs, respectively; α balances the contributions of intra-class compactness and inter-class separability.

Warm-up schedule. Applying a strong RS loss at the very beginning undermines the effect of $\mathcal{L}_{cos}^{(1)}$ and hinders

the adapter’s adaptation to the dataset (as evidenced in Sec. 5.3). Therefore, we first warm up the RS coefficient $\lambda_{RS}(e)$ so the adapter can learn a coarse understanding of the dataset, and then gradually increase the strength of representation steering. Specifically, the RS coefficient is defined as:

$$\lambda_{RS}(e) = \lambda_{RS} \cdot \min\left(1, \frac{e}{E_w}\right), \quad (5)$$

where λ_{RS} is its target value after warm-up; e is the current epoch index; E_w is the warm-up duration (in epochs); and $\min(1, e/E_w)$ is the linear warm-up factor that grows from 0 to 1 and then remains capped at 1 (so that $\lambda_{RS}(e) = \lambda_{RS}$ once $e \geq E_w$). The base task training loss can be written as:

$$\mathcal{L}^{(1)} = \mathcal{L}_{cos}^{(1)} + \lambda_{RS}(e) \mathcal{L}_{RS}. \quad (6)$$

4.2. Incremental Representation Steering

RAE alignment. Updating the shared adapter for task t inevitably perturbs the representation learned at task $t - 1$. Inspired by optimal transport theory [20, 31], we align the $t - 1$ and t spaces by minimizing an alignment loss with a learnable projector. Prior PTM-based CIL studies report that strong regularizers (e.g., knowledge distillation) can slow or even impede the adapter’s learning [38].

To avoid this problem while enforcing alignment, we adopt a residual autoencoder (RAE) that learns a nonlinear projection while retaining an identity skip path, as shown by the projector on the right side of Fig. 2. We fix the down-sampling dimension to 64 while adjusting the middle up-sampling dimension to control the number of trainable parameters. Let $f^{t-1} = \phi(x; A^{t-1})$ be the features from the frozen $(t - 1)$ -th model and $f^t = \phi(x; A^t)$ be the features

Table 1. Experimental results on six CIL datasets with **ViT-B/16-IN21K** as the backbone. We report average accuracy $\bar{A}$ over tasks and final accuracy A_B . ‘IN-R/A’ stands for ‘ImageNet-R/A’, and ‘OmniBench’ stands for ‘OmniBenchmark’. All methods are implemented without using exemplars. Best performance is shown in **bold**, and second-best performance is shown in underline.

Method	CIFAR <i>B0I10</i>		IN-R <i>B0I20</i>		IN-A <i>B0I20</i>		OmniBench <i>B0I30</i>		CUB <i>B0I10</i>		VTAB <i>B0I10</i>	
	$\bar{A}$	A_B	$\bar{A}$	A_B	$\bar{A}$	A_B	$\bar{A}$	A_B	$\bar{A}$	A_B	$\bar{A}$	A_B
L2P [44]	89.23	84.29	75.46	69.77	49.39	41.71	73.36	64.69	67.05	56.25	77.11	77.10
DualPrompt [43]	87.36	82.30	73.10	67.18	53.71	41.67	73.92	65.52	77.47	66.54	83.36	81.23
CODA-Prompt [35]	91.30	86.85	77.97	72.27	53.54	42.73	77.03	68.09	84.00	73.37	83.90	83.02
RanPAC [28]	94.35	90.86	82.98	77.94	69.32	61.82	<u>85.95</u>	79.55	93.13	89.40	92.56	91.83
SLCA [62]	94.09	91.53	81.17	77.00	68.66	58.74	82.80	74.10	89.51	82.19	90.94	90.76
APER+Adapter [66]	90.94	85.75	75.82	67.95	60.47	49.37	80.75	74.37	92.21	86.73	85.95	84.35
EASE [67]	92.35	87.76	81.74	76.17	65.34	55.04	81.11	74.85	92.23	86.81	93.61	<u>93.55</u>
MOS [37]	<u>94.75</u>	<u>91.70</u>	82.96	77.93	69.13	59.12	85.91	<u>80.05</u>	<u>93.49</u>	<u>90.12</u>	92.62	<u>92.79</u>
SSIAT [38]	94.35	91.35	<u>83.63</u>	<u>79.38</u>	<u>70.83</u>	<u>62.43</u>	84.31	<u>77.63</u>	93.38	89.53	<u>94.21</u>	91.28
RSIAT (Ours)	95.15	92.20	86.92	82.75	74.89	66.23	86.42	81.27	93.99	90.59	94.65	94.69

from the t -th model. The projector is defined as:

$$P^t(f^{t-1}) = \text{AutoEncoder}^t(f^{t-1}) + f^{t-1}. \quad (7)$$

Because the adapter and the projector are co-trained, the projector must continually track the model’s evolving representation. At the start of task t , we initialize $A^t = A^{t-1}$ and the projector to zero. Under the residual form, the projector is effectively an identity mapping (i.e., an I -projection), so the projection is perfectly aligned with the model at initialization. As optimization proceeds and the adapter updates, the projector only needs to model the incremental change of the representation on top of the identity path, it learns to account for the drift from f^{t-1} to f^t without imposing constraints on the adapter. This design lets the projector keep up with representation shifts while avoiding any hand-brake effect on the adapter’s parameter updates. As shown in Fig. 2, the projector projects f^{t-1} and aligns them with f^t . The L_2 alignment loss used can be expressed as:

$$\mathcal{L}_{\text{align}}^{(t)} = \frac{1}{|\mathcal{B}_t|} \sum_{x \in \mathcal{B}_t} \|f^t - P^t(f^{t-1})\|_2^2. \quad (8)$$

We use the $\mathcal{L}_{\text{align}}^{(t)}$ and RAE projector to align the gradually evolving feature space and avoid the optimization resistance caused by strong regularization.

Orthogonal representation steering. To reduce representational conflicts between the old and new classes, we use the memory of old class prototypes $\{\mu_c\}_{c \in \mathcal{Y}_{1:t-1}}$ accumulated at previous tasks. We map both the old prototypes and the old-tasks features into the current space with the projector $P^t(\cdot)$ and ℓ_2 -normalize them, $\hat{p}_c = P^t(\mu_c)/\|P^t(\mu_c)\|$ and $\hat{u}_i = P^t(f_i^{t-1})/\|P^t(f_i^{t-1})\|$. We then calculate the cosine similarity among these normalized vectors to encourage them to be orthogonal, thus ensuring that learning new classes does not affect the representation of old classes. The

orthogonal loss is expressed as follows:

$$\mathcal{L}_{\text{orth}}^{(t)} = \frac{1}{|\mathcal{Y}_{1:t-1}| |\mathcal{B}_t|} \sum_{c=1}^{|\mathcal{Y}_{1:t-1}|} \sum_{i=1}^{|\mathcal{B}_t|} |\langle \hat{p}_c, \hat{u}_i \rangle|, \quad (9)$$

where $|\langle \hat{p}_c, \hat{u}_i \rangle|$ represents the absolute value of the cosine similarity between the prototype and the feature. $\mathcal{L}_{\text{orth}}^{(t)}$ steers the separation of new classes from the prototypes of old classes, learning new classes with minimal impact on old classes, thus enhancing the model’s stability.

The incremental task training loss can be written as:

$$\mathcal{L}^{(t)} = \mathcal{L}_{\text{cos}}^{(t)} + \beta \mathcal{L}_{\text{align}}^{(t)} + \gamma \mathcal{L}_{\text{orth}}^{(t)}. \quad (10)$$

Inference procedure. During inference, the residual autoencoder is discarded. RSIAT infers with the frozen PTM backbone, the shared adapter, and the classifier, keeping the inference-time parameter count strictly constant. We summarize the pseudo code of RSIAT in Appendix A.

5. Experiments

We evaluate RSIAT on six CIL datasets against state-of-the-art methods under various CIL settings, and further verify its effectiveness with ablation studies, hyperparameter analyses, and t-SNE visualizations.

5.1. Implementation Details

Benchmark datasets. We evaluate the methods on six CIL benchmarks datasets, CIFAR100 [19], ImageNet-R [12], ImageNet-A [13], OmniBenchmark [63], CUB200 [39] and VTAB [59]. CIFAR100 contains 100 classes, ImageNet-R, ImageNet-A, and CUB200 contain 200 classes, OmniBenchmark contains 300 classes, and VTAB contains 50 classes. These six datasets have a large domain gap compared to the ImageNet dataset used for PTMs, making them suitable for use as evaluation datasets. Following the benchmark setting [67], we denote **BmIn** as m base classes and

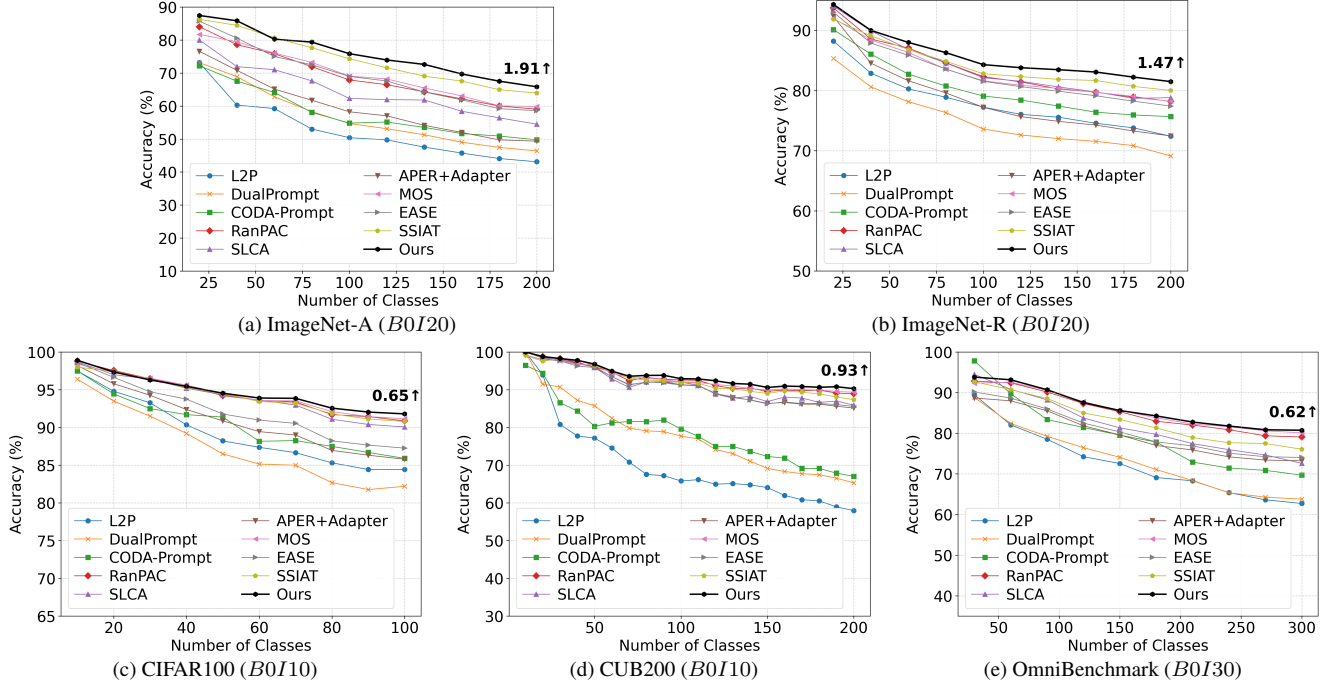


Figure 3. Performance curve on five CIL datasets with *ViT-B/16-IN1K* as the backbone. All methods are implemented without using exemplars. *BmIn* denotes *m* base classes and *n* new classes per incremental task. Annotate the final incremental task with numbers to indicate RSIAT’s relative improvement over the runner-up method.

n newly added classes per incremental tasks. Class orders are randomly shuffled with seed 1993, and all methods share the same train/test partitions. More details about the datasets can be found in the Appendix B.

Baseline methods. We engage both PTM-Based CIL methods and traditional CIL methods as our baselines. PTM-based CIL methods include L2P [44], DualPrompt [43], CODA-Prompt [35], RanPAC [28], SLCA [62], APER+Adapter [66], MOS [37], EASE [67], and SSIAT [38]. Traditional CIL methods include iCaRL [32], DER [47], FOSTER [40], MEMO [65], and TagFex [64]. Please refer to Appendix C for more details.

Training configurations. We run all experiments on NVIDIA 4090 and reproduce the compared methods with PyTorch [29] within the PILOT framework [36]. We use ViT-B/16 [6] as the PTM and consider two backbones: *ViT-B/16-IN21K* and *ViT-B/16-IN1K*. Both are initialized from ImageNet-21K pretraining, while the latter is further fine-tuned on ImageNet-1K. Following the work [38], we set the adapter dimension as 64 and use the cosineAnneal scheduler. We set different parameters for different datasets. The specific parameters are detailed in the Appendix D.

Evaluation metrics. We report two standard metrics: final accuracy [24] and average accuracy [32]. Given accuracies $a_{k,j} \in [0, 1]$ measured on the j -th task ($j \leq k$) after training tasks from 1 to k . The final accuracy, $A_B = \frac{1}{B} \sum_{j=1}^B a_{B,j}$, reflects the performance after learning B tasks. The average

Table 2. Comparison to traditional CIL methods with *ViT-B/16-IN21K* as the backbone. Baselines use 20 exemplars per class.

Method	Exemplars	IN-R <i>B0I20</i>		CIFAR <i>B0I10</i>	
		$\bar{A}$	A_B	$\bar{A}$	A_B
iCaRL [32]	20 / class	72.42	60.67	82.46	73.87
DER [47]	20 / class	80.48	74.32	86.04	77.93
FOSTER [40]	20 / class	81.34	74.48	89.87	84.91
MEMO [65]	20 / class	74.80	66.62	84.08	75.79
TagFex [64]	20 / class	83.23	78.45	92.17	89.26
RSIAT	0	86.92	82.75	95.15	92.20

accuracy, $\bar{A} = \frac{1}{B} \sum_{k=1}^B A_k = \frac{1}{B} \sum_{k=1}^B (\frac{1}{k} \sum_{j=1}^k a_{k,j})$, summarizes performance across all stages and captures how well the model maintains accuracy as classes accumulate. Higher A_B and $\bar{A}$ indicate superior continual learning.

5.2. Overall Evaluation

We compare RSIAT with PTM-based and traditional CIL methods using two backbone initializations across multiple incremental settings.

Tab. 1 reports results on six CIL datasets with *ViT-B/16-IN21K* as the pre-trained backbone [37]. RSIAT achieves the best average accuracy $\bar{A}$ and the best final accuracy A_B on all six datasets. The advantages are pronounced on ImageNet-A/R, indicating that RSIAT performs well across diverse data domains and exhibits strong robustness. Fig. 3 shows accuracy curves under the *ViT-B/16-IN1K* pre-trained backbone. RSIAT is competitive at early stages and pro-

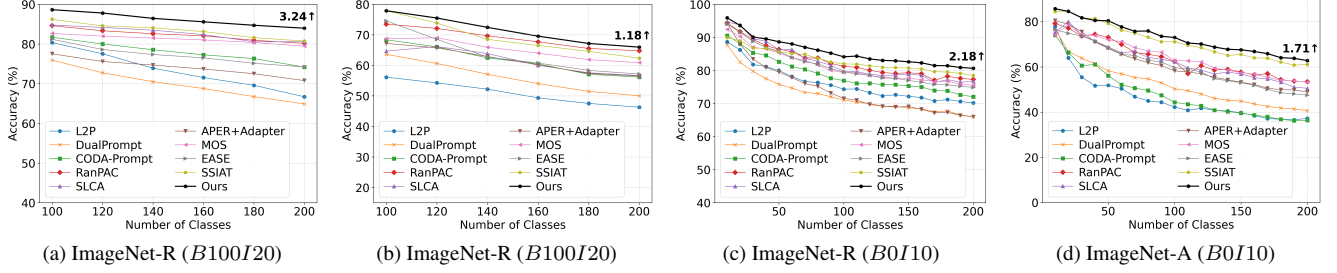


Figure 4. Results of IN-R/A in CIL settings with large base classes and long sequence tasks. All methods use the same backbone *ViT-B/16-IN21K*. Annotate the final incremental task with numbers to indicate RSIAT’s relative improvement over the runner-up method.

Table 3. Ablation study on ImageNet-R and ImageNet-A datasets.

Method	IN-R <i>B0I20</i>		IN-A <i>B0I20</i>	
	$\bar{A}$	A_B	$\bar{A}$	A_B
Baseline	83.63	79.38	70.83	62.43
+ $\mathcal{L}_{orth}$	84.15	78.32	72.83	61.95
+ $\mathcal{L}_{align}$	84.72	80.17	74.30	65.04
+ IRS	84.86	80.52	74.37	65.31
+ IRS (<i>wo_res</i>)	80.59	74.02	65.37	51.48
+ IRS + $\mathcal{L}_{RS}$	85.36	81.45	74.62	65.50
RSIAT	86.92	82.75	74.89	66.23

gressively widens its lead as classes accumulate, reflecting our designs: (i) shaping compact within-class structure and separated inter-class geometry at the base stage, and (ii) aligning spaces and encouraging separation between new and old features in later stages. The final-stage annotations mark our improvement over the runner-up method: +1.91 on ImageNet-A and +1.47 on ImageNet-R; other datasets also show improvements to varying degrees.

Following the setting in [42], we compare RSIAT with the traditional CIL methods combined with the *ViT-B/16-IN21K* backbone. It is worth noting that RSIAT is exemplar-free, whereas the engaged baselines are exemplar-based methods. As shown in Tab. 2, RSIAT is competitive even when the baselines engage 20 examples pre-class. Besides, we conduct large base classes and long sequence tasks on ImageNet-R/A by using *ViT-B/16-IN21K* as the pre-trained backbone. As shown in Fig. 4, RSIAT (i) starts competitively at early increments, (ii) degrades more gracefully as the number of classes grows, and (iii) achieves the highest last-stage accuracy. It confirms that RSIAT adapts well to diverse experimental settings. More experimental results are provided in the Appendix E.1 & E.2.

5.3. Ablation Study

We ablate each RSIAT component on ImageNet-R and ImageNet-A in Tab. 3. Starting from the Baseline (cosine head and shared adapter only), independently adding the orthogonal loss (+ $\mathcal{L}_{orth}$) improves average accuracy ($\bar{A}$) by preventing new features from overlapping with old prototypes, though it slightly restricts final plasticity (A_B). Al-

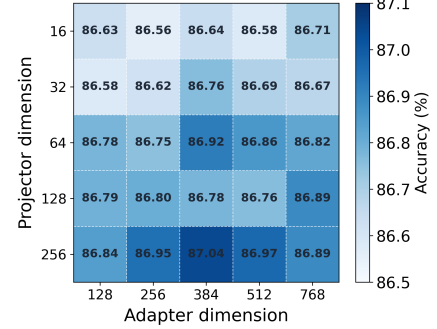


Figure 5. Hyperparameter study on ImageNet-R (*B0I20*). Heatmap of average accuracy $\bar{A}$ versus projector dimension d_p (x-axis) and adapter bottleneck d_a (y-axis).

ternatively, introducing the alignment loss (+ $\mathcal{L}_{align}$) aligns the current embedding space with the projected latent space to explicitly mitigate feature drift, yielding consistent gains across both metrics. Combining them into Incremental Representation Steering (+ IRS) provides the optimal synergy, effectively keeping old and new concepts separated. The + IRS (*wo_res*) variant removes the projector’s residual identity path, causing a severe performance drop (e.g., A_B falls to 51.48% on IN-A). This confirms that without the identity prior, the projector over-regularizes the adapter and hinders the acquisition of new knowledge. Integrating Base Representation Steering (+ IRS + $\mathcal{L}_{RS}$) brings further gains, proving that a compact base feature space seeds stable downstream adaptation. Finally, the full RSIAT introduces a warm-up schedule. The notable improvement over the non-warmed-up variant validates the need to initially carve margins before tightening clusters, ultimately achieving the most favorable stability-plasticity trade-off.

5.4. Further Analysis

Hyperparameter study. RSIAT contains two hyperparameters that affect the number of trainable parameters: the adapter bottleneck dimension d_a and the residual projector code dimension d_p . We sweep $d_a \in \{16, 32, 64, 128, 256\}$ and $d_p \in \{128, 256, 384, 512, 768\}$ on ImageNet-R (*B0I20*) and report the average accuracy $\bar{A}$ in Fig. 5. The results show that the performance surface is relatively flat around a ridge near $d_p \approx$

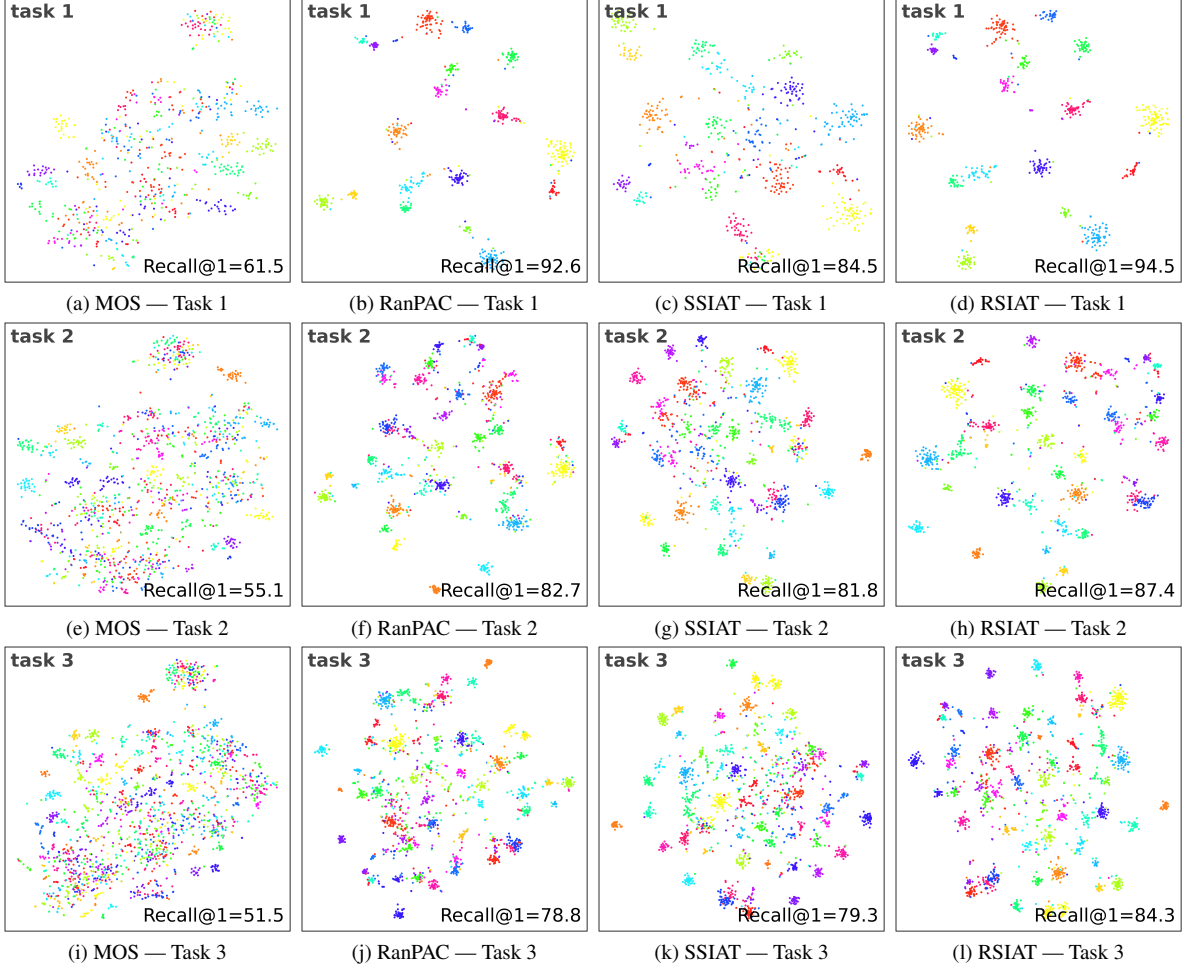


Figure 6. t-SNE embeddings across tasks and methods. Columns correspond to Task 1/2/3, rows correspond to MOS, RanPAC, SSIAT, and RSIAT (ours). Numbers in each panel denote Recall@1 measured in the original feature space before t-SNE. RSIAT forms compact clusters and preserves separation across tasks.

384 and $d_a \in [64, 256]$. Although the absolute peak occurs at $(d_p=384, d_a=256)$ with $\bar{A} \approx 87.04$, using $(d_p=384, d_a=64)$ yields $\bar{A} \approx 86.92$ (only ~ 0.12 lower) while reducing the adapter parameter count to roughly 1/4. Given the small performance gap, we adopt $(d_p=384, d_a=64)$ as the default to minimize extra parameters with minimal accuracy sacrifice.

Visualizations. We analyze how different methods shape the feature space across three incremental tasks (Task 1/2/3) under the *ViT-B/16-IN21K* backbone. At each task, we visualize the features with t-SNE [25] and report the nearest-neighbor class-match rate Recall@1 [48] in the original feature space to quantify within-class compactness and between-class dispersion. As shown in Fig. 6, RSIAT produces the most compact and well-separated clusters at all three tasks (Task 1/2/3) and achieves the highest and most stable Recall@1 scores ($94.5\% \rightarrow 87.4\% \rightarrow 84.3\%$), with a markedly smaller drop than MOS, RanPAC, and SSIAT. It indicates that RSIAT learns and maintains strong represen-

tations (tighter within-class cohesion and larger between-class margins) while exhibiting smaller cross-task prototype drift. More t-SNE results are given in the Appendix E.3.

6. Conclusion

We introduced Representation-Steered Incremental Adapter Tuning (RSIAT), a unified framework for parameter-efficient class-incremental learning. By steering representation learning in the base task and employing a residual autoencoder to align features across tasks, RSIAT achieves consistent and discriminative representations without increasing model parameters. This design effectively balances stability and plasticity, eliminating the linear parameter growth common in prior PTM-based approaches. Extensive experiments across diverse benchmarks validate RSIAT’s efficiency and robustness, offering a scalable solution for continual adaptation in large pre-trained models. Future work will explore broader modalities and further structural guidance for lifelong representation learning.

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